

Some physical aspects of Positron Emission Tomography

A walk through Ganguly et al., J. Radioanal. Nucl. Chem. 279 (2009)

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Outline

1. Foundations of PET
2. From β^+ decay to annihilation
3. Detection and image reconstruction
4. Radiotracers, limits and conclusions

Foundations of PET

Molecular imaging technique mapping a 3D *biological function* (vs. CT/MRI = anatomy).

The basic idea, in five steps:

1. Label a biomolecule with a β^+ -emitter (e.g. ^{18}F -FDG).
2. Inject and let it distribute biologically.
3. Each e^+ annihilates $\rightarrow$ two **511 keV** photons **back-to-back**.
4. Detector ring catches the pairs **in coincidence** $\rightarrow$ each pair = one Line of Response (LOR).
5. Millions of LORs $\rightarrow$ tomographic reconstruction $\rightarrow$ 3D image.

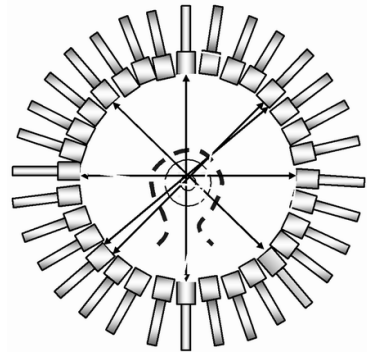
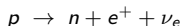


Figure 1: Ring of PET detectors, true coincidence LORs [Ganguly et al., 2009].

The source: β^+ decay

- A proton-rich nucleus converts a proton into a neutron, emitting a positron and a neutrino:



Energetic threshold

- β^+ decay requires $Q > 2m_e c^2 = 1.022 \text{ MeV}$;
- below threshold, only **EC** is energetically allowed.

Practical requirements

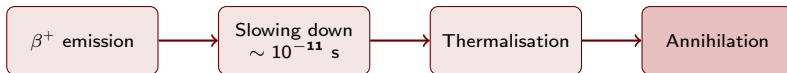
- short $T_{1/2} \Rightarrow$ **cyclotron on-site** (or generator for ^{68}Ga , ^{82}Rb);
- pure β^+ emitters preferred: extra γ would create false coincidences.

Common PET isotopes

Isotope	$T_{1/2}$ (min)	$E_{\max}$ (MeV)	Production
^{18}F	109.7	0.63	$^{18}\text{O}(p, n)^{18}\text{F}$
^{11}C	20.3	0.96	$^{14}\text{N}(p, \alpha)^{11}\text{C}$
^{15}O	2.07	1.73	$^{14}\text{N}(d, n)^{15}\text{O}$
^{68}Ga	67.7	1.90	^{68}Ge generator
^{82}Rb	1.27	3.15	^{82}Sr generator

The positron in matter & positronium

Positron path in matter:



The annihilation occurs in two possible ways:

Path 1 (~ 70%)

Direct annihilation: the e^+ meets a free e^- of the medium $\Rightarrow 2\gamma$.

Path 2 (~ 30%)

Positronium formation: e^+ binds with a tissue e^- to form **positronium** (Ps):

- **para-Ps** ($S = 0$): $\tau \approx 125$ ps $\rightarrow 2\gamma$;
- **ortho-Ps** ($S = 1$): $\tau \approx 142$ ns $\rightarrow 3\gamma$.

Problem: 3γ from ortho-Ps would give no usable PET coincidence. How can PET still work?

Pick-off quenching of ortho-positronium

Why this matters: PET requires **two photons in coincidence**. But ortho-Ps would naturally decay into $3\gamma \Rightarrow$ no usable coincidence.

In vacuum: ortho-Ps $\rightarrow 3\gamma$

- Bound state: $e^+(\uparrow) + e^-(\uparrow)$, $S = 1$
- Lifetime: $\tau \approx 142$ ns
- Decay channel: 3γ
- No back-to-back pair $\Rightarrow$ no PET signal.

In tissue: pick-off

- The e^+ captures an **external $e^-(\downarrow)$** from a nearby atom before the 3γ decay can occur.
- The new pair is effectively a singlet $\Rightarrow 2\gamma$ back-to-back.
- Lifetime drops to ~ 1 -3 ns.
- The original e^- partner is left as a free electron.

$\Rightarrow$ PET detects 2 photons in coincidence regardless of the intermediate Ps state.

Annihilation: 511 keV and back-to-back photons

Conservation laws



Two conserved quantities fix the kinematics:

- **Energy** (pair at rest):

$$E_{tot} = 2m_e c^2 = 1.022 \text{ MeV}$$

$\Rightarrow E_\gamma = 511 \text{ keV}$ per photon.

- **Momentum**: photons
back-to-back (180°).

Non-collinearity: an intrinsic limit

- Residual momentum of e^+e^- pair
($\sim \text{keV}/c$) $\Rightarrow$ deviation from 180° ,
FWHM $\sim 0.5^\circ$.

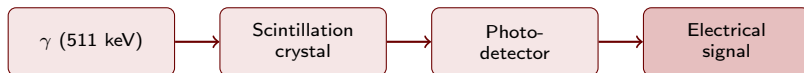
- Position uncertainty on the LOR:

$$\Delta R \approx \frac{D}{2} \tan(0.25^\circ)$$

- $D = 80 \text{ cm} \Rightarrow \Delta R \approx 3.5 \text{ mm} \Rightarrow$
1.5–2 mm FWHM contribution to PET
resolution.
- **Cannot be eliminated**; only mitigated by
smaller scanners (e.g. brain/breast PET).

Radiation detectors: scintillators + photodetectors

Detection chain:



Scintillation crystal (LSO, LYSO, BGO...)

- High ρ , $Z \rightarrow$ stops the photon
- High light yield $\rightarrow$ good energy resolution
- Short decay time $\rightarrow$ good time resolution (TOF)
- Not hygroscopic $\rightarrow$ stable, easy to build

Trade-off: BGO has highest Z but slow; LaBr_3 is fastest but expensive.

Photodetector

Amplifies $\sim$ thousand light photons to a measurable pulse:

- **PMT**: vacuum tube, dynode chain $\Rightarrow$ gain 10^6 – 10^7 .
- **SiPM**: solid-state, compact, MR-compatible, modern standard.

Energy resolution:

E.R. = FWHM/centroid $\sim 12\%$ (LSO).
Tight rejection of Compton-scattered γ via energy window (350-650 keV).

PET instrumentation: electronic collimation & sinogram

Electronic collimation: two opposite photons in coincidence define the LOR *automatically*.

An event is valid if:

1. within *coincidence window* ($\sim 3\text{--}4$ ns;
 $\Delta t_{\max} = D/c$);
2. LOR within *acceptance angle*;
3. energies within *energy window*.

Full 360° ring $\Rightarrow$ complete angular sampling for tomography.

The sinogram $p(s, \phi, \theta, z)$

4D histogram: each cell counts the coincidences for one LOR, identified by 4 parameters:

- s : perp. distance to FOV centre;
- ϕ : azimuthal angle;
- θ : polar angle (3D PET);
- z : axial position.

Input of the reconstruction algorithm.

Image reconstruction: from sinogram to 3D image

Each detector pair sees the integral of the tracer along its **tube of response**:

$$\langle \mathcal{P}_{D_A D_B} \rangle = \tau \int_{\text{FOV}} d\vec{r} f(\vec{r}) \zeta_{D_A D_B}(\vec{r})$$

Goal: recover $f(\vec{r})$ from millions of LOR measurements.

Two families of algorithms

- **FBP:** analytic Radon inversion. Fast, noisy.
- **Iterative (MLEM, OSEM):** statistical, slower, higher quality. *Today's clinical standard.*

Time-of-Flight (TOF) PET

Δt between photons $\rightarrow$ event localised on LOR with

$$\sigma_x = \frac{c \sigma_t}{2}$$

$\sigma_t = 200 \text{ ps} \Rightarrow \sigma_x \approx 3 \text{ cm}.$

Sensitivity gain $\sim 5\times$ for large patients.

Radiotracers and clinical applications

PET isotopes (^{11}C , ^{13}N , ^{15}O , ^{18}F) coincide with the building blocks of organic chemistry $\Rightarrow$ **virtually any biomolecule can be labelled.**

Common tracers

- ^{18}F -FDG: oncology, brain
- ^{15}O - H_2O : cerebral blood flow
- ^{11}C -raclopride: dopamine receptors
- ^{68}Ga -DOTA-peptides: neuroendocrine tumours
- ^{11}C -PiB: amyloid plaques (Alzheimer)

Why ^{18}F -FDG is the workhorse

- Low $E_{\text{max}} = 0.63 \text{ MeV} \rightarrow$ short range.
- $T_{1/2} = 110 \text{ min} \rightarrow$ ship-able.
- Cancer cells consume glucose anomalously $\rightarrow$ high contrast.

Critical points: the physical limits of PET

Intrinsic limits (positron physics)

- **Positron range** (0.5–5 mm): PET sees the *annihilation site*, not the decay site.
- **Non-collinearity** (1.5–2 mm): residual momentum of e^+e^- pair.

Detector-side limits

- Crystal size (intrinsic detector resolution).
- Parallax error.
- Scatter and random coincidences.

Modern solutions

- **TOF-PET** → event localised on LOR ⇒ less noise.
- **DOI detectors** → encode depth of interaction ⇒ kill parallax.
- **PET/MR** (with SiPM) → MR-compatible imaging hybrid.
- **Total-body PET** → $\sim 40\times$ sensitivity.

Thank you for your attention!

Questions?

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Backup: Radionuclide generators (^{68}Ge , ^{82}Sr)

Some PET isotopes are produced **not in a cyclotron, but via a chemical column** loaded with a long-lived parent isotope. The daughter (the actual PET tracer) is "milked" by elution.

$^{68}\text{Ge}/^{68}\text{Ga}$ generator

- Parent: $^{68}\text{Ge} \rightarrow T_{1/2} = 271$ days (EC decay)
- Daughter: $^{68}\text{Ga} \rightarrow T_{1/2} = 68$ min (PET tracer)
- Generator lifetime: ~ 1 year

$^{82}\text{Sr}/^{82}\text{Rb}$ generator

- Parent: $^{82}\text{Sr} \rightarrow T_{1/2} = 25$ days (EC decay)
- Daughter: $^{82}\text{Rb} \rightarrow T_{1/2} = 75$ s (cardiac PET)
- Generator lifetime: $\sim 4\text{--}6$ weeks

How it works

1. Parent isotope is produced in a (distant) cyclotron and loaded onto an ion-exchange column.
2. Column is shipped to the hospital.
3. Daughter accumulates slowly via parent decay.
4. When needed, the column is **eluted** with a saline solution: the daughter is washed out, the parent stays bound.

Why it matters

Enables PET imaging in clinical sites **without a cyclotron on-site**. Crucial for the diffusion of PET in smaller hospitals.

Backup: Why $\sim 70\%$ direct vs $\sim 30\%$ positronium?

The 70/30 ratio is **empirical** (measured in water and biological tissues), *not* a fundamental constant.

Competing processes (end of positron track)

At thermal energies, the positron either:

- **Annihilates directly** with an electron of the medium ($\sim 70\%$)
- **Captures an electron** from its own ionization track to form positronium ($\sim 30\%$)

The Spur Model (Mogensen, 1974)

At the end of its path, the positron sits in a **spur** — a cluster of molecules it has just ionized, with a few quasi-free electrons around. Ps formation requires the positron to capture one of these electrons *before* they recombine.

Medium dependence

Ps fraction varies strongly with chemistry:

- Water / biological tissue: $\sim 30\%$
- Liquid organic solvents: 30–50%
- Polymers: 10–50%
- Liquid argon: $\sim 50\%$
- Metals (delocalised e^-): $\sim 0\%$
- Vacuum (free e^- available): $\sim 100\%$

Bottom line

The number reflects the *chemistry of the medium*, not a property of the positron itself.